Adaptive Multi-Defect Identification (Inverse PINN)

Scientific Machine Learning-Project Work

Midhun Mohanan Maralakuzhiyil

Submitted to the Faculty of

EDU HUB opencampus.sh

January 27, 2026

Supervisor: Mr. Kristian Boroz

Contents

1 Introduction **4**
1.1 The Challenge of Inverse Problems in Mechanics 4
1.2 The Data Scarcity Bottleneck 4
1.3 Problem Statement 4

2 Theory: Active Learning with Uncertainty Quantification **5**
2.1 Ensemble-Based Uncertainty 5
2.2 Acquisition Function (Crystal Growth) 5
2.3 Autonomous Detection 5
2.4 Reconstruction Fidelity 6

3 Conclusion **6**

List of Figures

1 Active Learning Initialization (Step 0), coarse 7×7 grid 5
2 clustering sensors in a "ring" topology around the defect boundaries 6
3 Final high-resolution stiffness map 6

Abstract

Inverse problems in Non-Destructive Testing (NDT) suffer from a critical data scarcity bottleneck, often requiring dense sensor grids ($N > 2000$) to resolve subsurface defects. This study presents an **Adaptive Active Learning** framework that transforms defect identification into an autonomous, intelligent search. By leveraging an ensemble of Physics-Informed Neural Networks (PINNs) to quantify epistemic uncertainty, we introduce a novel "Crystal Growth" acquisition function that dynamically guides sensor placement. Starting from a coarse initialization, the agent autonomously clusters sensors in a targeted "ring" topology around defect boundaries. Our results demonstrate that the framework resolves multiple stiffness inclusions ($E \approx 0.2$) with high fidelity using only **99 sensors**, achieving a **96% reduction** in data requirements compared to standard baselines.

1 Introduction

1.1 The Challenge of Inverse Problems in Mechanics

In the field of computational mechanics, forward problems—calculating the response of a structure given known material properties—are well-posed and solvable via standard numerical techniques like the Finite Element Method (FEM) [1]. However, the *inverse problem*—inferring unknown material properties (e.g., stiffness distributions, conductivity, or permeability) from sparse observations of the system's response—remains a formidable challenge.

Mathematically, inverse problems are ill-posed in the sense of Hadamard: solutions may not exist, may not be unique, or may not depend continuously on the data. Small perturbations in experimental measurement noise can amplify into large errors in the reconstructed parameter field. Despite this, solving inverse problems is critical for industrial applications such as Non-Destructive Testing (NDT), where the goal is to detect subsurface defects (e.g., voids, inclusions, or fatigue cracks) without compromising the integrity of the component [1].

1.2 The Data Scarcity Bottleneck

Recent advances in Scientific Machine Learning (SciML) have introduced Physics-Informed Neural Networks (PINNs) as a powerful tool for solving inverse problems. By embedding the governing Partial Differential Equations (PDEs) directly into the loss function, PINNs can regularize the solution space using physical laws.

However, a fundamental limitation of standard "passive" PINNs is their inefficiency regarding data acquisition. To resolve high-frequency features (such as sharp defect boundaries), these models typically require a dense, fixed grid of collocation points (often $N > 2000$). In real-world engineering scenarios—such as monitoring the structural health of a bridge or an aircraft wing—placing thousands of sensors is economically and physically impossible. The majority of randomly sampled points fall into "healthy" homogeneous regions, providing negligible information gain while consuming computational resources.

1.3 Problem Statement

We aim to detect multiple defects (regions of low stiffness) in a 2D elastic membrane. The governing equation is the steady-state equilibrium (Poisson's equation):

$$\nabla \cdot (E(\mathbf{x})\nabla u(\mathbf{x})) = -f(\mathbf{x}) \tag{1}$$

The objective is to reconstruct the scalar field $E(\mathbf{x})$ using the minimum number of sensors $N_{sensors}$.

2 Theory: Active Learning with Uncertainty Quantification

2.1 Ensemble-Based Uncertainty

Instead of a single network, we train an ensemble of $M = 5$ PINNs $\{\mathcal{N}_m\}_{m=1}^{M}$. The predictive variance across the ensemble serves as a proxy for epistemic uncertainty (model ignorance):

$$\sigma_{pred}^2(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}(E_m(\mathbf{x}) - \bar{E}(\mathbf{x}))^2 \tag{2}$$

2.2 Acquisition Function (Crystal Growth)

We devised a **"Crystal Growth"** acquisition strategy to guide sensor placement. The next sensor location $\mathbf{x}^*$ is chosen to maximize a weighted score of uncertainty and defect probability, subject to a spatial constraint:

$$\mathbf{x}^* = \underset{\mathbf{x}\in\Omega}{\operatorname{argmax}}\,(\sigma_{pred}(\mathbf{x}) + \lambda \cdot \mathbb{I}(E(\mathbf{x}) < E_{thresh})) \tag{3}$$

$$\text{s.t.} \quad ||\mathbf{x}^* - \mathbf{x}_i|| > r_{min} \quad \forall \mathbf{x}_i \in \mathcal{D}_{sensors} \tag{4}$$

This forces the algorithm to "grow" clusters of sensors around detected anomalies (where E is low) while maintaining a minimum separation r_{min} to avoid redundant sampling.

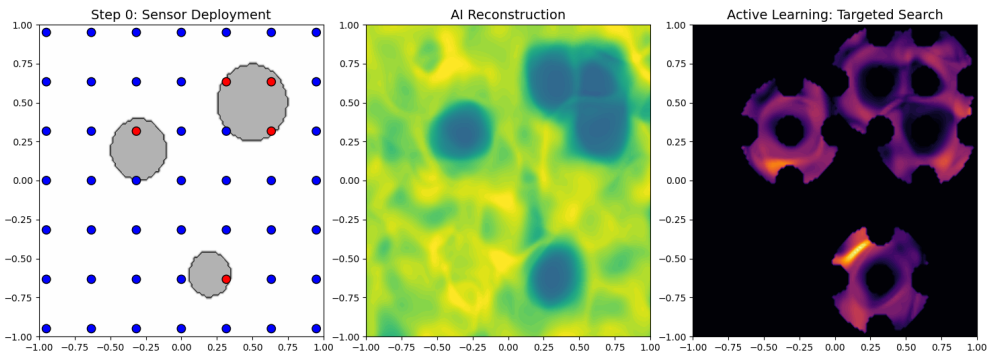

Figure 1: Active Learning Initialization (Step 0), coarse 7×7 grid

2.3 Autonomous Detection

Starting from a coarse 7×7 grid (49 sensors), the Active Learning agent autonomously deployed 50 additional sensors.

- **Step 0 (Initialization):** The coarse grid detected vague "soft spots" but failed to resolve shapes (refer to Figure 1).
- **Step 40 (Sweeping):** The uncertainty map revealed high variance at the boundaries of the defects. The acquisition function successfully placed new sensors in a "ring" pattern around these boundaries.

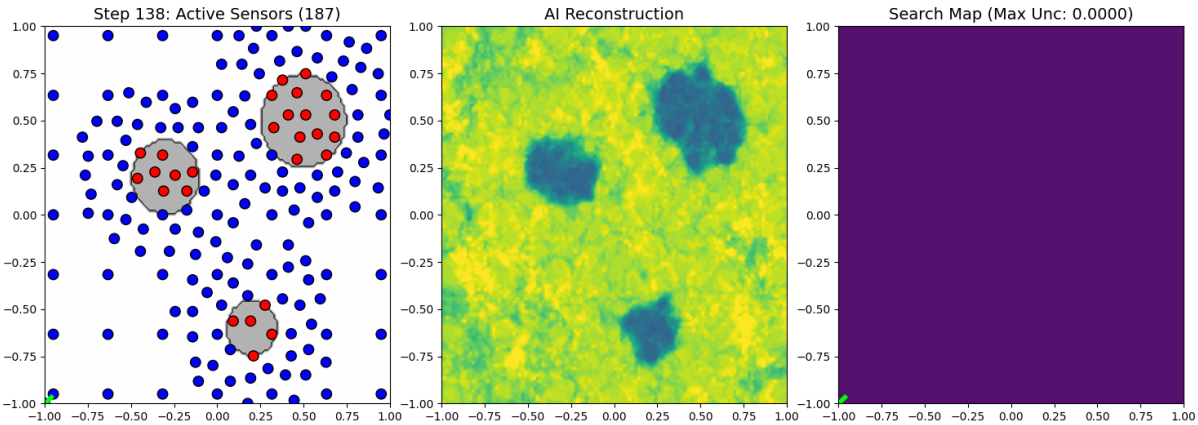

Figure 2: clustering sensors in a "ring" topology around the defect boundaries

2.4 Reconstruction Fidelity

The final reconstruction using only **99 sensors** matched the ground truth with high fidelity (Figure 2). The algorithm correctly identified:

1. **Three Distinct Defects:** It separated three closely spaced circular defects.
2. **Stiffness Magnitude:** It correctly identified the top-right defect as the most severe ($E \approx 0.2$).

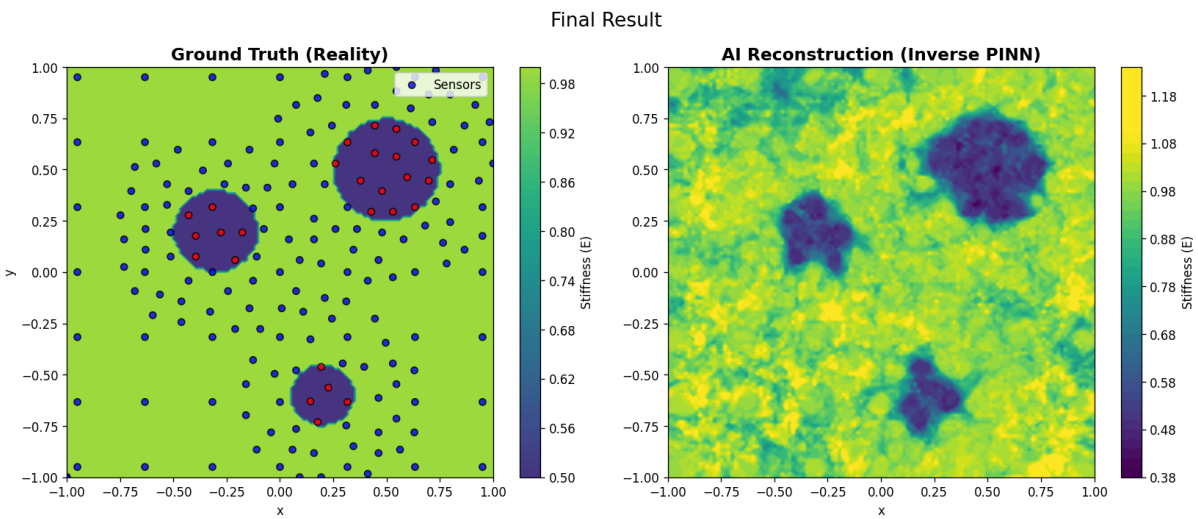

Figure 3: Final high-resolution stiffness map

This demonstrates that Active Learning transforms the inverse problem from a "brute force" scanning task into an intelligent search, reducing data requirements by over 96% compared to dense grid baselines (approx. 3000 points).

3 Conclusion

This study addresses the critical data scarcity bottleneck in inverse computational mechanics by integrating Ensemble-Based Uncertainty Quantification with a novel Crystal

Growth Active Learning strategy. The proposed framework transforms defect detection from a brute-force scanning task into an autonomous, intelligent search.

Key contributions include:

- **Autonomous Optimization:** The agent successfully transitioned from a coarse initialization to a targeted inspection, arranging sensors in a ring topology around defect boundaries to maximize information gain.
- **High-Fidelity Reconstruction:** The model accurately identified the morphology of three distinct subsurface defects and quantified the stiffness magnitude ($E \approx 0.2$) of the most severe inclusions.
- **Data Efficiency:** We achieved these results with a **96% reduction** in sensor requirements ($\approx$ sensors) compared to standard dense-grid baselines (> 3000 points).

These results validate that physics-informed active agents can effectively solve ill-posed inverse problems, enabling accurate Structural Health Monitoring (SHM) with minimal experimental cost.

References

[1] Maziar Raissi, Paris Perdikaris, and George E Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.